ML Project–
Out-Of-Distribution Plant Disease Identification

Due: 11:59 pm 27/11/2026 (Friday of Week 12)
Contributes 50% of your final result
Group Assignment: Group of 4-5 students

1 Introduction

Out-of-Distribution (OOD) Data refers to data that differs significantly from the data used to train a machine learning model. When a model is trained on a specific set of classes or objects, it learns to identify and classify those objects based on their features. However, when the model encounters an object that was not part of its training data, this new object is considered out-of-distribution.

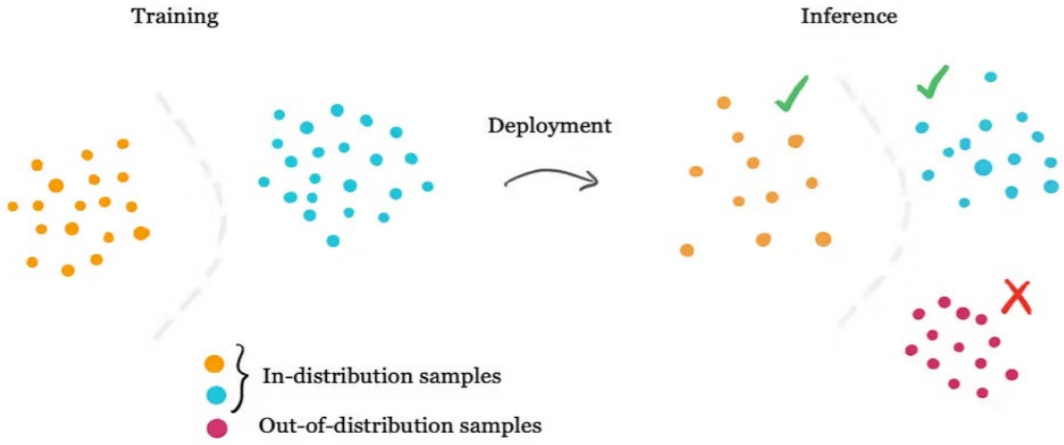


Fig. 1 Out of Distribution Samples

Zero-Shot Object Identification involves recognizing and identifying objects that the model has not seen during its training phase. This is achieved by leveraging auxiliary information, such as semantic embeddings, descriptions, or relationships between known and unknown classes. In essence, the model is tasked with generalizing its knowledge to recognize new, unseen objects.

Key Points:

- Training vs. Test Data: In zero-shot learning, the training data consists of a certain set of objects (in-distribution), while the test data includes objects that the model has never seen (out-of-distribution).
- Semantic Information: Zero-shot learning often relies on semantic information, such as word embeddings or attributes, to bridge the gap between seen and unseen classes. This auxiliary information helps the model to infer the characteristics of the unseen objects based on their relationship with the seen objects.
- Generalization: The ability of a zero-shot model to correctly identify OOD objects depends on its generalization capability, which is enhanced by the use of robust semantic embeddings and a well-designed training process.

2. Unseen Plant Disease Identification

Plant diseases significantly affect agricultural productivity, food security, and ecosystem sustainability. Accurate and timely identification of plant diseases is essential for effective crop management and disease control. Traditionally, plant disease diagnosis relies on expert knowledge to analyse visual symptoms, such as leaf lesions, spots, colour variations, and morphological changes. However, expert-based diagnosis is often costly, time-consuming, and difficult to scale, particularly in large-scale agricultural systems and regions with limited access to specialists.

Recent advances in artificial intelligence (AI), particularly deep learning and computer vision, have enabled automated plant disease identification with high accuracy. Nevertheless, most existing AI-based approaches assume a closed-world setting, where all disease categories encountered during deployment are known during training. This assumption limits their practical application because agricultural environments are dynamic, and new diseases, emerging variants, or previously unobserved crop–disease relationships may appear over time.

This project focuses on the challenge of **unseen plant disease identification**, where an AI model is expected to recognise disease categories that are not available during training. The central research challenge is to achieve a balance between **discriminative ability** and **generalisation capability**. A discriminative model should capture disease-specific characteristics to distinguish visually similar diseases, while a generalisable model should learn transferable representations that enable recognition beyond the observed training categories. Developing models that achieve both objectives remains a fundamental challenge in machine learning.

The formulation of unseen plant disease identification itself is an open research question. A disease category may involve different levels of abstraction, including crop characteristics, disease symptoms, biological knowledge, visual patterns, environmental factors, or other semantic relationships. For example, a model may consider disease recognition from a **compositional perspective** by decomposing categories into meaningful concepts such as crop type and disease type. Alternatively, other formulations may be explored depending on how the model represents and transfers knowledge between seen and unseen categories.

Let C denote the complete set of disease categories, where the training categories are represented as S and the unseen categories are represented as U:

$$C = S \cup U$$

The training data only provides information from categories in S, while the objective is to develop a model capable of recognising categories in U. Students are required to investigate and design an appropriate formulation and learning strategy that enables effective knowledge transfer from seen to unseen plant disease categories.

Students are encouraged to critically analyse the relationship between representation learning, category definition, and generalisation ability. Possible directions may include, but are not limited to, compositional learning, vision-language approaches, knowledge-guided learning, feature disentanglement, domain adaptation, generative approaches, or other emerging AI methodologies. The selected approach should be justified based on its ability to address the challenges of unseen plant disease identification.

3. Getting Started

In this project, students are required to investigate and develop deep learning approaches for unseen plant disease identification. The project consists of two main components: (1) establishing baseline approaches and (2) proposing a new approach to improve the baseline performance.

3.1 Baseline Approach

The baseline approaches are designed to evaluate the capability of conventional convolutional neural networks (CNNs) in learning transferable representations from seen disease categories. An overview of the baseline framework is shown in Fig. 1.



Fig. 1: Baseline approach for unseen plant disease identification

Students are required to implement and evaluate CNN-based baseline models using images from the seen domain (S). Specifically, students should investigate the following baseline strategies:

1. **Independent CNN models (Left in Fig.1)**: Train two separate CNN models to predict the two target concepts, namely the crop category and disease category, respectively.
2. **Multi-task CNN model (Right in Fig.1):** Train a single CNN backbone with a multi-task learning framework, where the model simultaneously learns crop and disease classification objectives.

Any state-of-the-art CNN architecture (e.g., ViT, ResNet, EfficientNet, ConvNeXt, or other suitable architectures) may be adopted as the backbone model. Students are expected to apply techniques covered in this course, including but not limited to data preprocessing, data augmentation, transfer learning, and model optimisation.

The baseline models serve as an important benchmark for evaluating whether conventional CNN-based approaches can learn meaningful and transferable visual representations from seen plant disease categories and generalise to unseen categories.

3.2 Proposed New Approach

Beyond the baseline models, students are required to propose and develop a new deep learning-based approach to improve unseen plant disease identification performance. The proposed approach should aim to address the fundamental challenges of **generalisation and robustness** in real-world plant disease recognition.

The proposed model design should consider the following research questions:

- **How can the model improve generalisation to recognise unseen plant disease categories that are not available during training?**

- **How can the model improve robustness when applied to real-world field images with variations in lighting conditions, backgrounds, viewpoints, and image quality, compared with controlled laboratory images?**

Students are encouraged to explore advanced deep learning techniques for learning discriminative and transferable representations. Possible directions include, but are not limited to:

- self-supervised learning,
- contrastive learning,
- vision-language models,
- knowledge-guided learning,
- feature disentanglement,
- domain adaptation,
- generative data augmentation,
- or other emerging AI approaches.

The proposed approach should be clearly motivated, technically justified, and experimentally evaluated. At a minimum, students must implement **one new deep learning-based approach** and compare its performance against the baseline models using the provided evaluation protocol.

3.3 System Evaluation

The evaluation metric used is Top-N accuracy, harmonic mean and confusion matrix. Top-N accuracy is a performance metric commonly used in machine learning, especially in classification tasks where the model has to predict a label from a set of possible labels. Top-N accuracy considers whether the true label is within the top N predictions made by the model. If the true label is within the top N predictions, it counts as correct for this metric. The harmonic mean is a type of average particularly when dealing with rates or ratios. The harmonic mean gives more weight to smaller values. This makes it particularly useful in scenarios where extremely large or small values can significantly impact the overall average. Confusion matrix provides a detailed breakdown of the model's predictions compared to the actual outcomes, allowing you to understand where the model is making errors.

$$Top - N\ accuracy = \frac{True\ positive + True\ negative}{True\ positive + False\ positive + True\ negative + False\ negative}$$

$$Harmonic\ mean = \frac{2 \times (seen\ accuracy) \times (unseen\ accuracy)}{(seen\ accuracy) + (unseen\ accuracy)}$$

- You are required to provide both Top-1 and Top-5 accuracy results.
- You are required to provide harmonic mean result between **seen** and **unseen** accuracies of DPD-Small dataset.
- You are required to provide the confusion matrix for seen testing images from DPD-Small dataset

4. Dataset

4.1 Plant Disease Dataset

This project will utilise the recently proposed Deep-plant-disease (DPD) dataset, which is one of the largest and most comprehensive plant disease image datasets developed for large-scale plant disease

recognition research. The dataset contains diverse crop species and disease categories, providing a challenging benchmark for evaluating the generalisation capability of AI models.

However, considering the computational resource requirements and storage limitations typically available for student projects, a reduced version of the dataset, namely **DPD-Small**, has been curated specifically for this project. **DPD-Small** maintains the essential characteristics and challenges of the original dataset while providing a more practical scale for model development, training, and evaluation within the available project constraints.

Students are required to conduct experiments and benchmarking using the DPD-Small dataset rather than the full DPD dataset. The objective is to develop and evaluate AI models capable of addressing the challenges of unseen plant disease identification while considering practical limitations in computational efficiency and resource usage.

The following section provides a detailed description of the dataset.

4.2 DPD-Small Dataset

The datasets and metadata can be downloaded here.

- DPD-Small dataset
- DPD-Small train.csv
- DPD-Small test seen.csv
- DPD-Small test unseen.csv

In this project, we utilize two distinct datasets: the Deep-Plant-Disease Small (DPD-Small) dataset. The DPD-Small dataset comprises images that represent real-world conditions. An overview of the dataset is provided below:

Dataset	**No. of images**			
	Train	**Test**		**Total**
		Seen	**Unseen**	
DPD-Small	8 000	2 000	1 448	11 448

File/Folder	**Content**
DPD-Small train.csv	List of training images with ground truth from the DPD-Small dataset
DPD-Small test seen.csv	List of seen testing images with ground truth from the DPD-Small dataset

DPD-Small test unseen.csv	List of unseen testing images with ground truth from the DPD-Small dataset

NOTE:

- Each metadata file (csv file) contains seven columns: the first column represents the image file name, the second column specifies the ground truth for the crop label category, the third column provides the ground truth for the disease label category, the fifth column specifies the ground truth for the crop name category, and the sixth column specifies the ground truth for the disease name category. The crop and disease categories are combined to form a single composition in the label form (fourth column) and the name form (seventh column).
- There are 55 crop categories, 175 disease categories and 323 seen crop-disease composition categories. The unseen consists of 10 crop-disease composition categories. The index starts from 0.

4. User Interface for plant disease identification system

The expected input to the system is the plant disease image and the expected output is to predict which crop-disease class it belongs to and display the relevant training images it matches. The design of the graphical interface should be user-friendly and practical.

Marking Scheme

Requirements	***Mark***
3.1 Model Implementation (25%): Implementation of the baseline approaches (12%) and the proposed new approach (13%).	25
3.2 Project Report (10%): This report comprises two sections: *Methodology* and *Results and Discussion*. It should provide a detailed description of the overall framework. The report presents comprehensive details of each model, encompassing information such as the training scheme, loss function, hyperparameters, and other relevant aspects. Furthermore, the report discusses the performance differences observed between the models and justifies the selection of the model that yielded the best results. Please limit the report to **8** pages.	10
3.3 User Interface (5%) The user interface requirements for the plant disease identification system are listed in Section 6.	5
3.4 Project Presentation (10%): You are required to present your project findings to the unit lecturer and industry partners. Please *record your video presentation and include the link in your report*. The video should be *no longer than 8 minutes*. A Q&A session will be scheduled, and the date and time will be announced on Canvas.	10
Total	**50**

NOTE:

- Individual marks will be proportionally adjusted based on each team member's overall contribution to the project as indicated in the '*Who did what*' declaration.
- You must also provide to shlee@swinburne.edu.my read only access to your git repository within 1 week of forming teams.

SUBMISSION:

- You must submit your *project report* to Canvas under ML Project submission's link by 11:59pm on 27/11/2026(Friday of Week 12). Standard late penalties apply - 10% for each day late, more than 5 days late is 0%.
    - Project report (max 8 pages) and provide link to your source code (Git-based VCS)).
- The *checklist*, *cover sheet* and *'Who did what'* declaration form, you can download them in Canvas